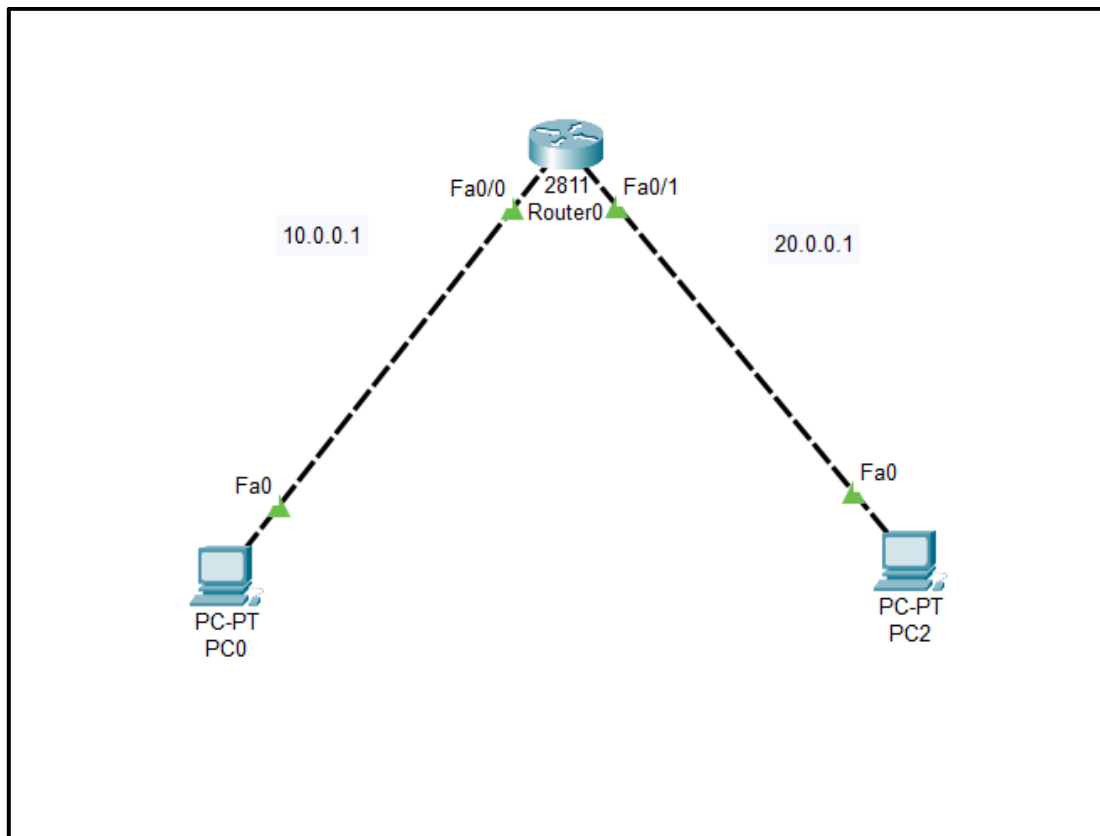


LAN



Would you like to enter the initial configuration dialog? [yes/no]: **no**

Press RETURN to get started!

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 20.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#**exit**

Router(config)#**exit**

Router#

%SYS-5-CONFIG_I: Configured from console by console

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FEC4:98BC

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 20.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 20.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE35:80D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Request timed out.

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

Reply from 20.0.0.2: bytes=32 time=1ms TTL=127

Ping statistics for 20.0.0.2:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

Reply from 20.0.0.2: bytes=32 time<1ms TTL=127

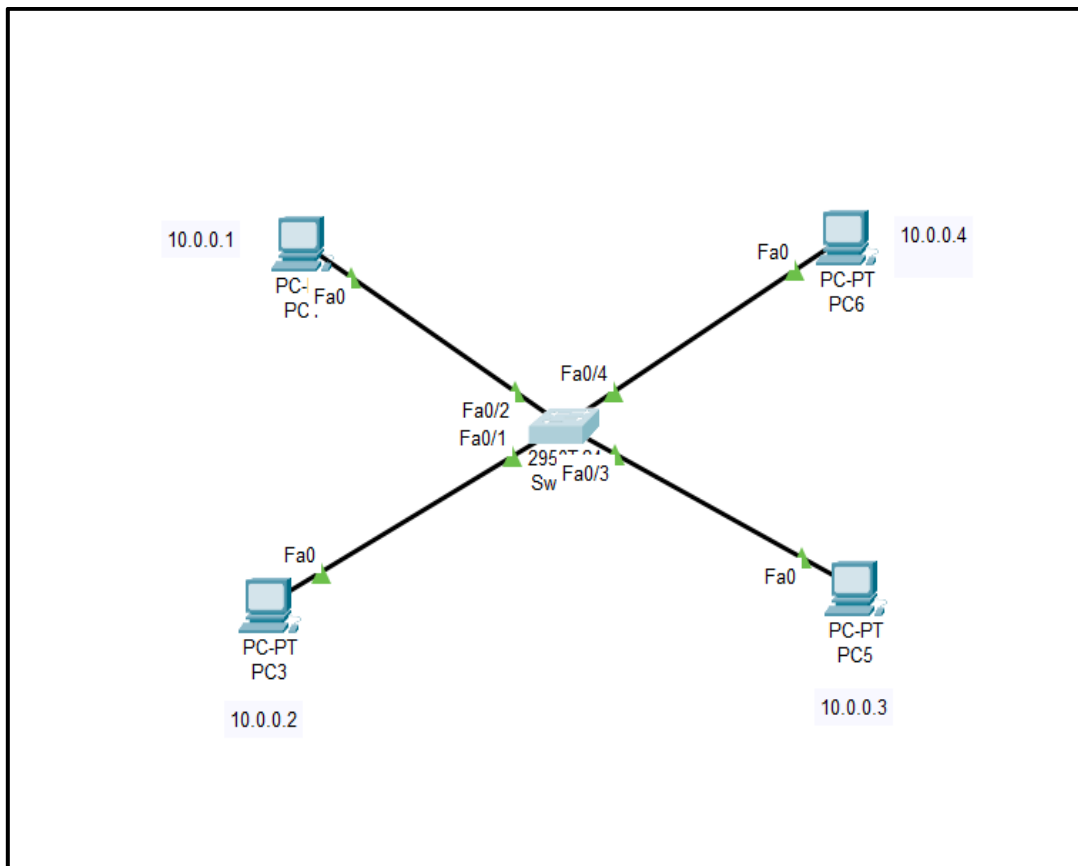
Ping statistics for 20.0.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

VLAN



Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 10

Switch(config-vlan)#exit

Switch(config)#interface fastethernet 0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#interface fastethernet 0/2

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#exit

Switch(config)#vlan 20

Switch(config-vlan)#exit

Switch(config)#interface fastethernet 0/3

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#interface fastethernet 0/4

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#exit

Switch#

%SYS-5-CONFIG_I: Configured from console by console

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FE3B:970A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.1

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:47FF:FEEA:5AB6

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Close

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE22:76C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC6

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.4

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:BFF:FE88:436

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=1ms TTL=128

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

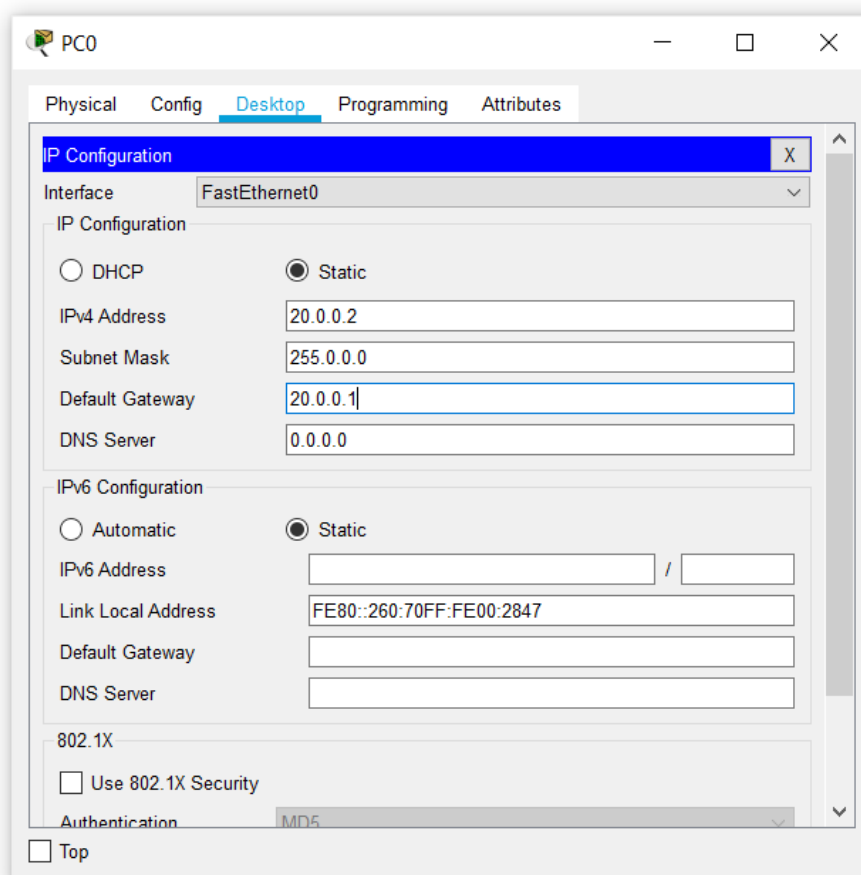
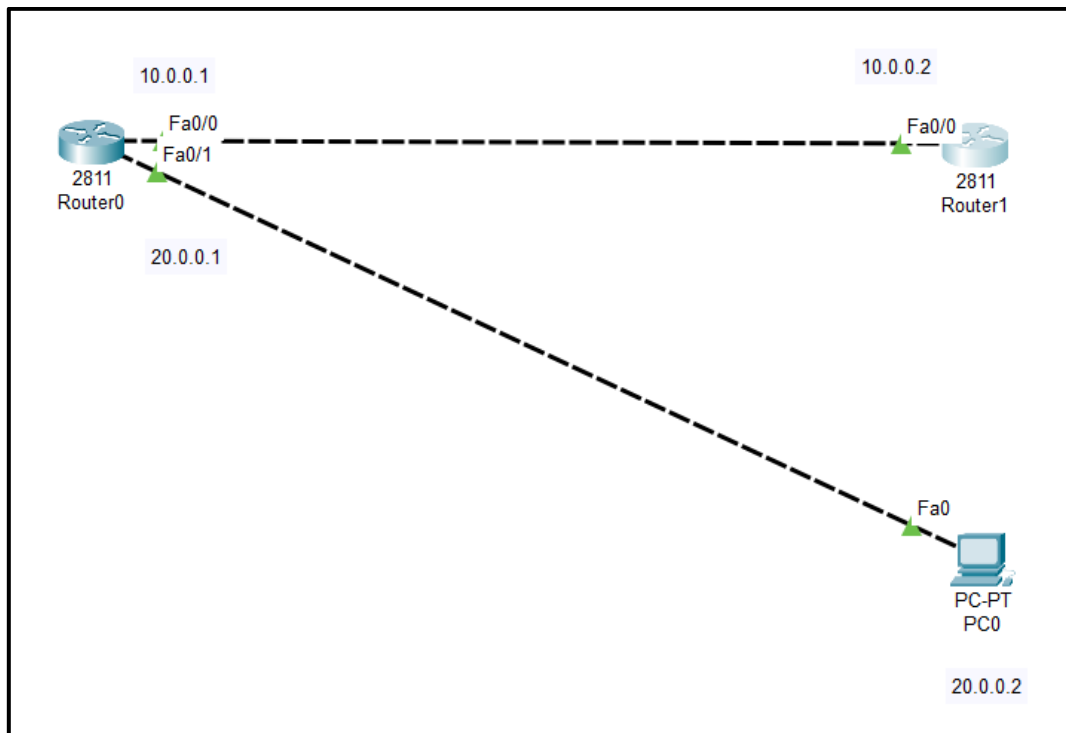
Minimum = 0ms, Maximum = 1ms, Average = 0ms

Request timed out.

Ping statistics for 10.0.0.4:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

TELNET



Router-0

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#interface fastethernet 0/1

Router(config-if)#ip address 20.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#exit

```
Router(config)#line vty ?
<0-15> First Line number
Router(config)#line vty 0 5
Router(config-line)#login local
Router(config-line)#transport input telnet
Router(config-line)#exit
Router(config)#username abc
Router(config)#username abc password 123
Router(config)#enable password 456
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
```

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router#[telnet 10.0.0.1](#)

Trying 10.0.0.1 ...Open

User Access Verification

Username: [abc](#)

Password:

Router>[enable](#)

Password:

Router#[configure terminal](#)

Enter configuration commands, one per line. End with CNTL/Z.

Changing Hostname-

Router-1

Router(config)#[hostname qwerty](#)

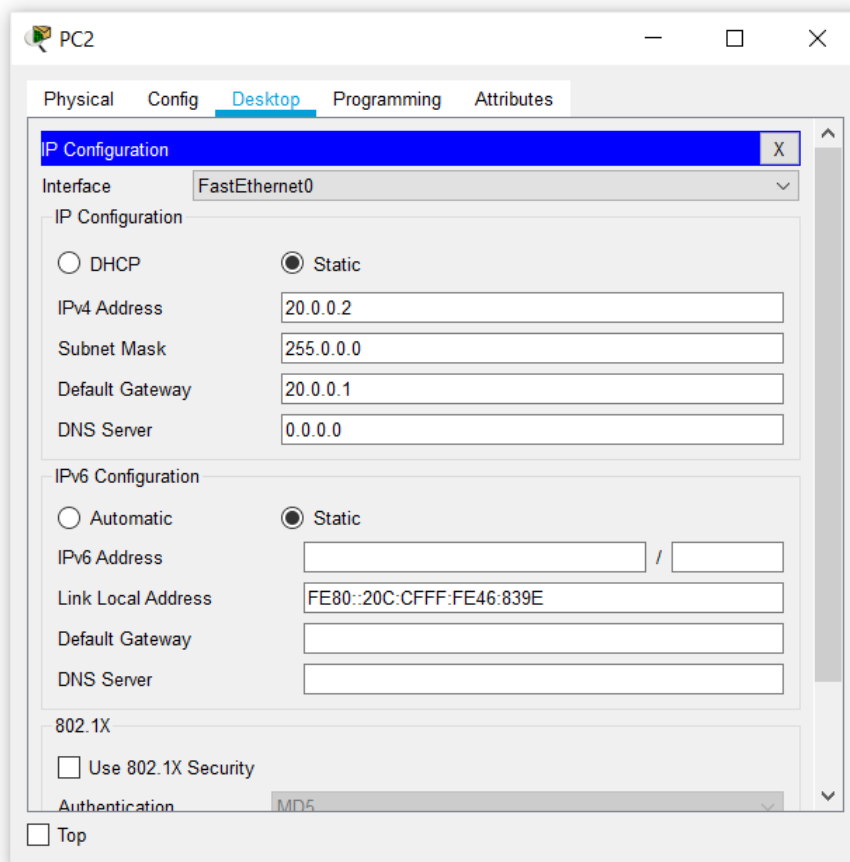
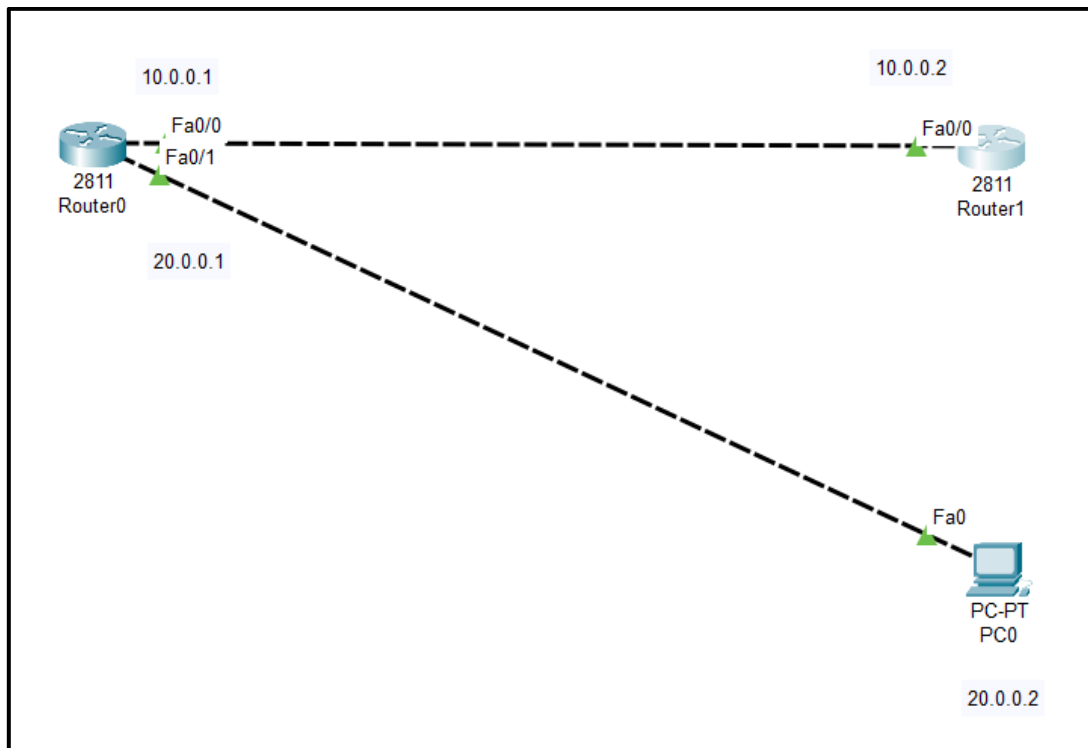
qwerty(config)#

Router-0

Router#

qwerty#

SSH



Router-0

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 20.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#**exit**

Router(config)#line vty ?

<0-15> First Line number

Router(config)#line vty 0 5

Router(config-line)#login local

Router(config-line)#transport input ssh

Router(config-line)#exit

Router(config)#username abc

Router(config)#username abc password 123

Router(config)#enable password 456

Router(config)#ip domain-name jkl

Router(config)#hostname qwerty

qwerty(config)#crypto key generate rsa

The name for the keys will be: qwerty.jkl

Choose the size of the key modulus in the range of 360 to 2048 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.

How many bits in the modulus [512]:

% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

qwerty(config)#exit

*Mar 1 0:8:23.437: RSA key size needs to be at least 768 bits for ssh version 2

*Mar 1 0:8:23.437: %SSH-5-ENABLED: SSH 1.5 has been enabled

qwerty#

%SYS-5-CONFIG_I: Configured from console by console

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 10.0.0.2 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

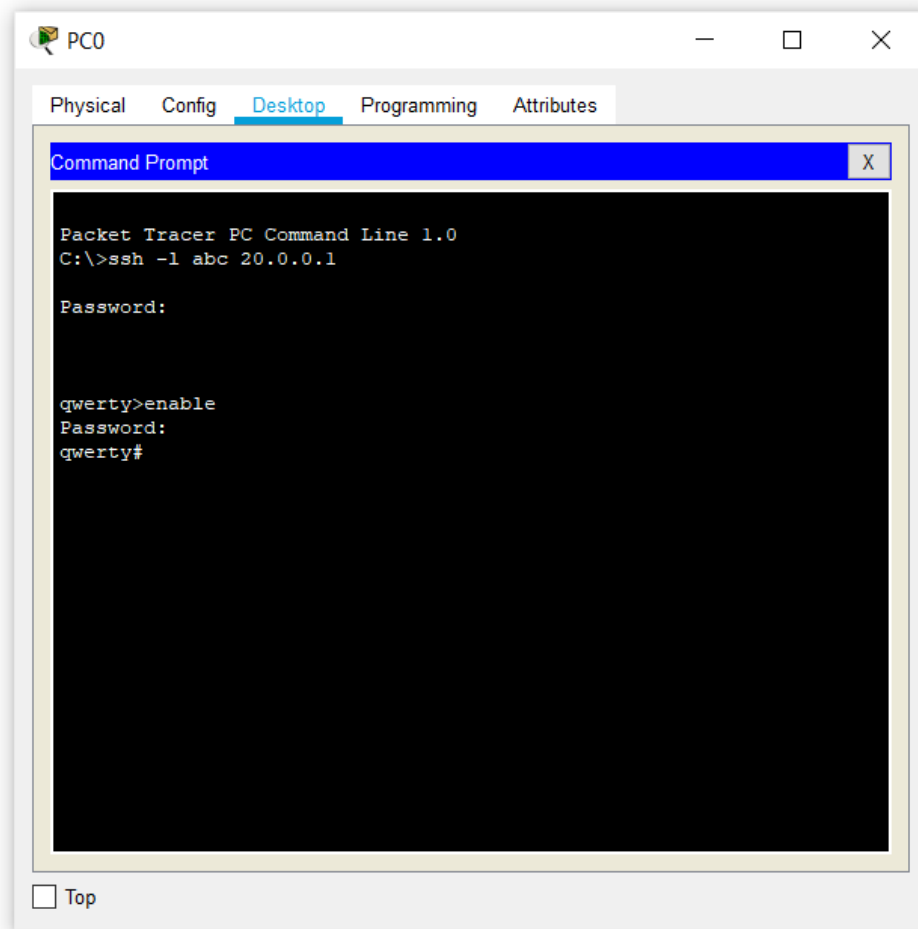
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#exit

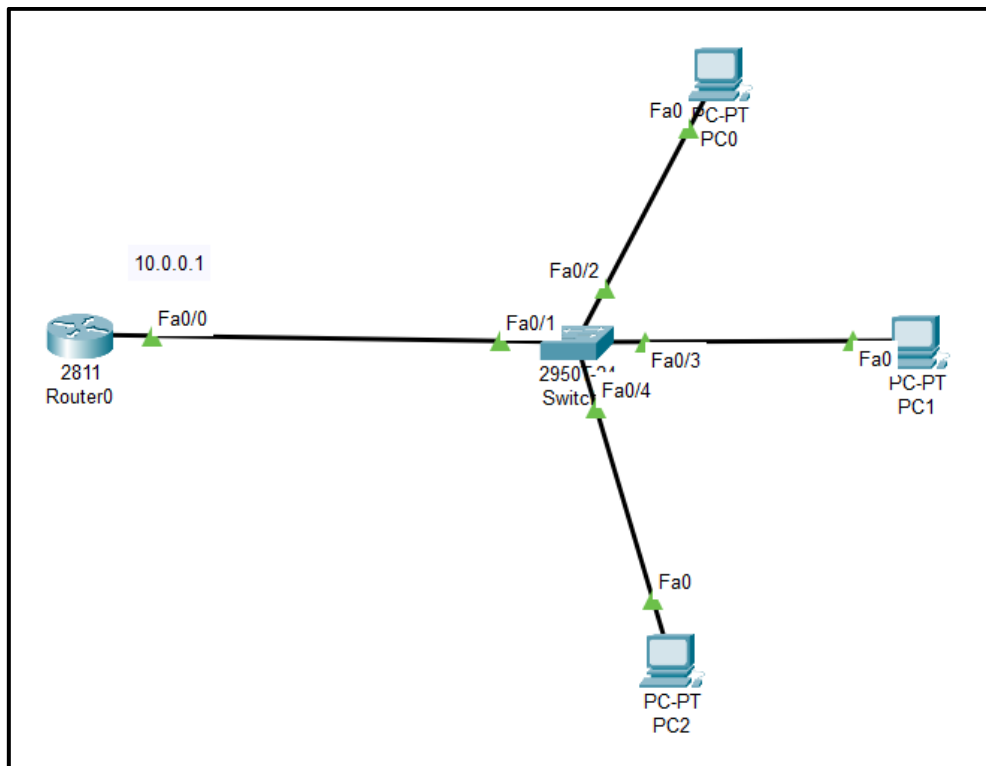
Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console



DHCP



Would you like to enter the initial configuration dialog? [yes/no]: **no**

```
Router>enable
```

```
Router#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#interface fastethernet 0/0
```

```
Router(config-if)#ip address 10.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#ip dhcp pool abc

Router(dhcp-config)#network 10.0.0.0 255.0.0.0

Router(dhcp-config)#default-router 10.0.0.1

Router(dhcp-config)#dns-server 20.20.20.20

Router(dhcp-config)#%DHCPD-4-PING_CONFLICT: DHCP address conflict:
server pinged 10.0.0.1.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 20.20.20.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::200:CFF:FE82:1D60

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 10.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 20.20.20.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::250:FFF:FEDB:641

Default Gateway

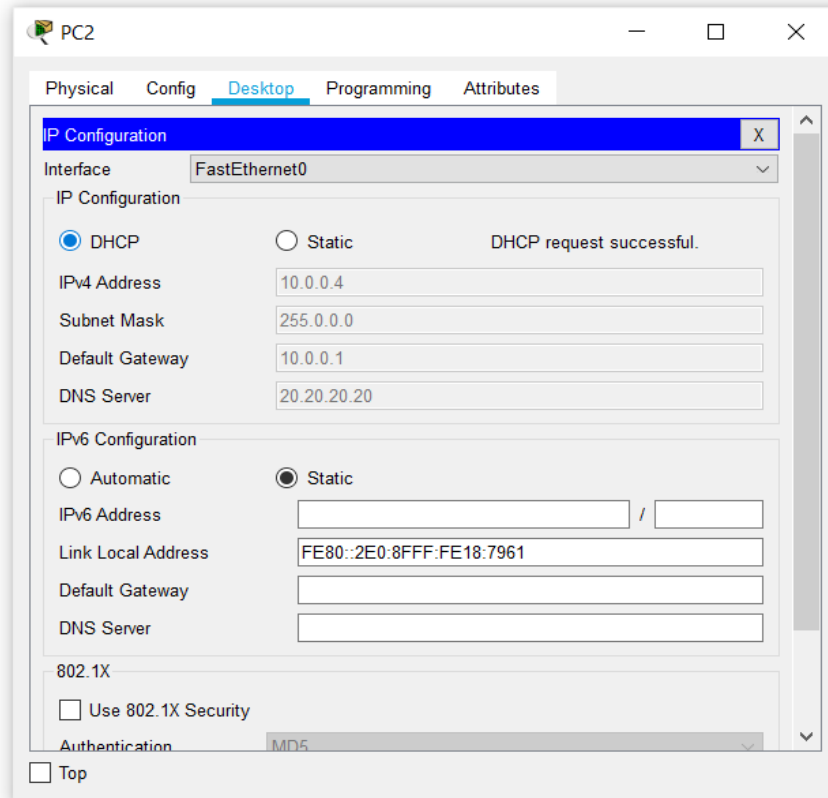
DNS Server

802.1X

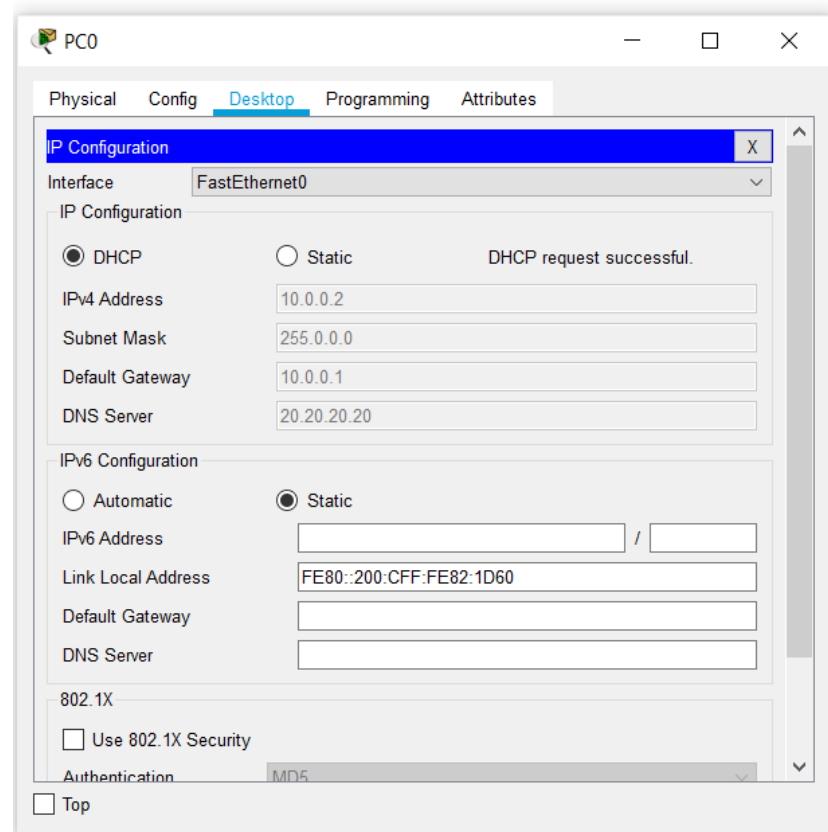
☐ Use 802.1X Security

Authentication MD5

☐ Top



➤ Router(dhcp-config)#ip dhcp excluded 10.0.0.3 10.0.0.25



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 10.0.0.27

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 20.20.20.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::250:FFF:FEDB:641

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 10.0.0.26

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 20.20.20.20

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:8FFF:FE18:7961

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

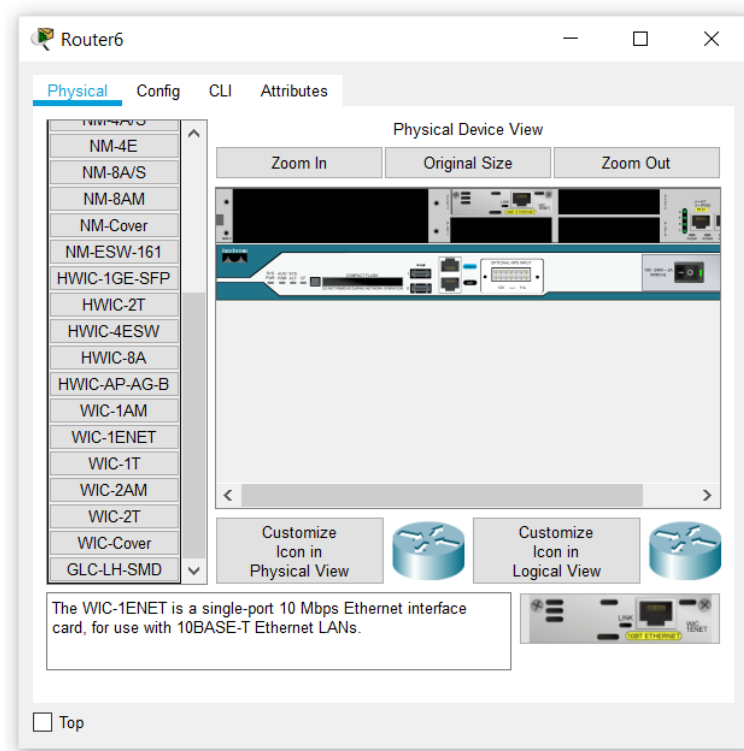
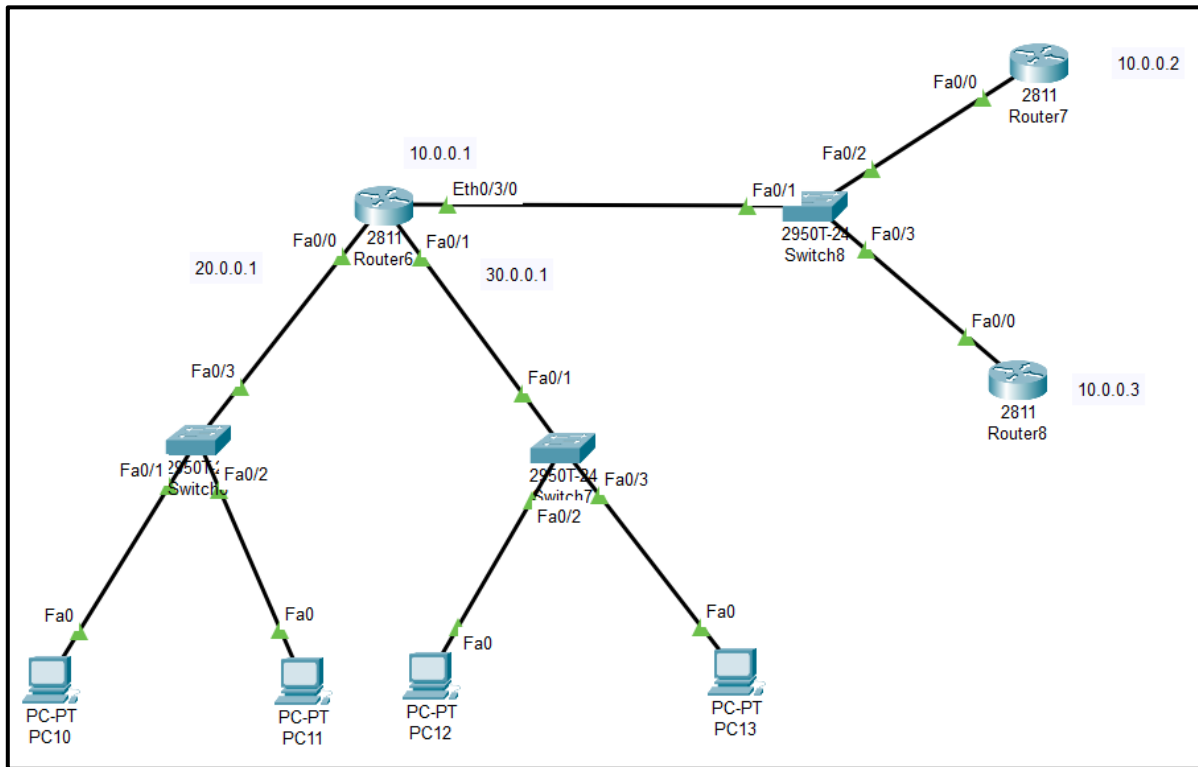
Authentication MD5

☐ Top

DHCP

Providing IP to PC-10, PC-11 Through R-7 &

Providing IP to PC-12, PC-13 Through R-8



Router-6

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 20.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#interface fastethernet 0/1

Router(config-if)#ip address 30.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#exit

```
Router(config)#interface ethernet 0/3/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface Ethernet0/3/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3/0,
changed state to up
```

```
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.3
Router(config)#interface fastethernet 0/0
Router(config-if)#ip helper 10.0.0.2
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet 0/1
Router(config-if)#ip helper 10.0.0.3
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Router-7

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,  
changed state to up
```

```
Router(config-if)#exit
```

```
Router(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

```
Router(config)#ip dhcp pool abc
```

```
Router(dhcp-config)#network 20.0.0.0 255.0.0.0
```

```
Router(dhcp-config)#default-router 20.0.0.1
```

```
Router(dhcp-config)#exit
```

```
Router(config)#
```

Router-8

```
Would you like to enter the initial configuration dialog? [yes/no]: no
```

```
Router>enable
```

```
Router#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Router(config)#interface fastEthernet 0/0
```

```
Router(config-if)#ip address 10.0.0.3 255.0.0.0
```

```
Router(config-if)#no shutdown
```

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#[exit](#)

Router(config)#[ip route 0.0.0.0 0.0.0.0 10.0.0.1](#)

Router(config)#[ip dhcp pool abc](#)

Router(dhcp-config)#[network 30.0.0.0 255.0.0.0](#)

Router(dhcp-config)#[default-router 30.0.0.1](#)

Router(dhcp-config)#[exit](#)

Router(config)#

PC11

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 20.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 20.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::230:F2FF:FEA3:81E

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC12

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::200:CFF:FEDA:B05C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC10

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 20.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 20.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FECD:87A7

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC13

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 30.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:B0FF:FEA0:2B42

Default Gateway

DNS Server

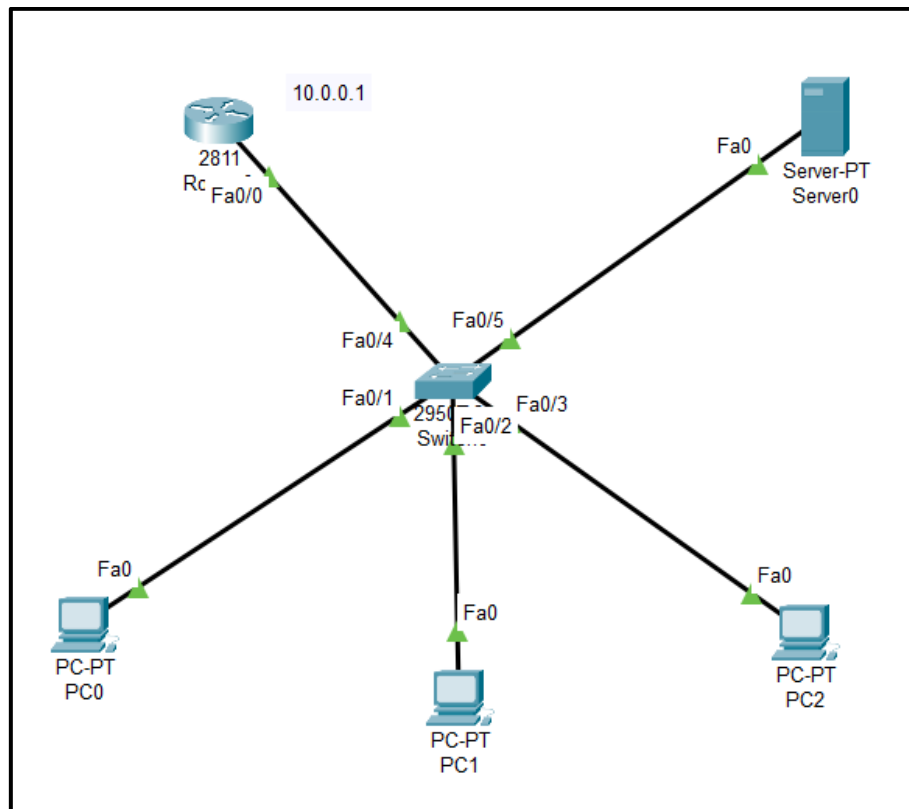
802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

DNS



Router-0

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

```
Router(config-if)#exit
```

```
Router(config)#ip dhcp pool abc
```

```
Router(dhcp-config)#network 10.0.0.0 255.0.0.0
```

```
Router(dhcp-config)#exit
```

```
Router(config)#
```

➤ Steps to follow in server after providing DHCP to all the PC-

#Server- Services- DNS- On

#HTTP- index.html- Edit The Content- Save

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type ARecord

Address

AddSaveRemove

No.	Name	Type	Detail
-----	------	------	--------

DNS Cache

☐ Top

Server0

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

HTTP

HTTP ☒ On ☐ Off

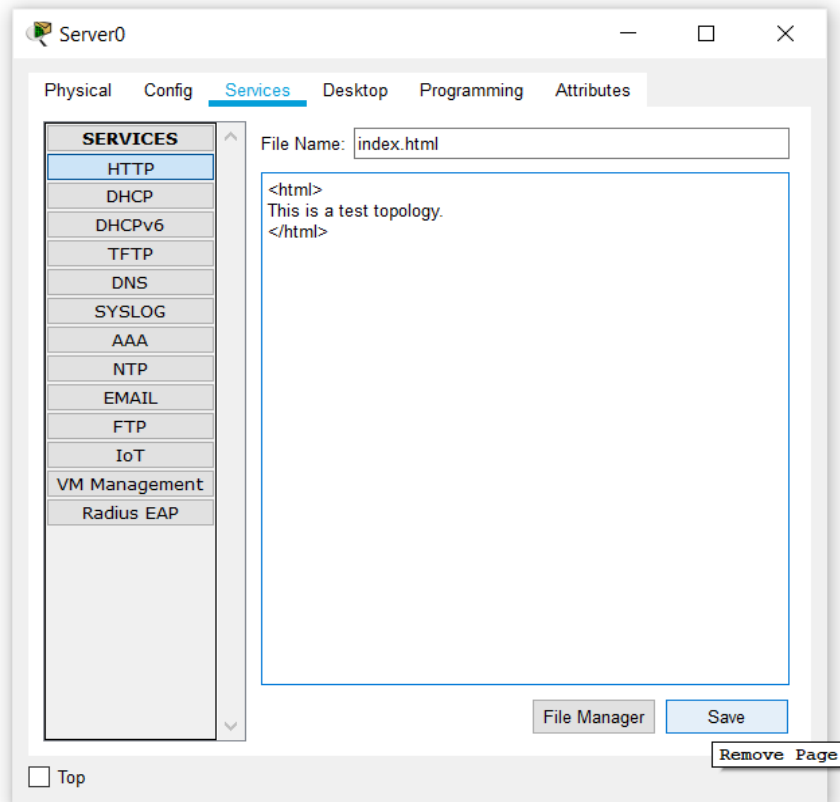
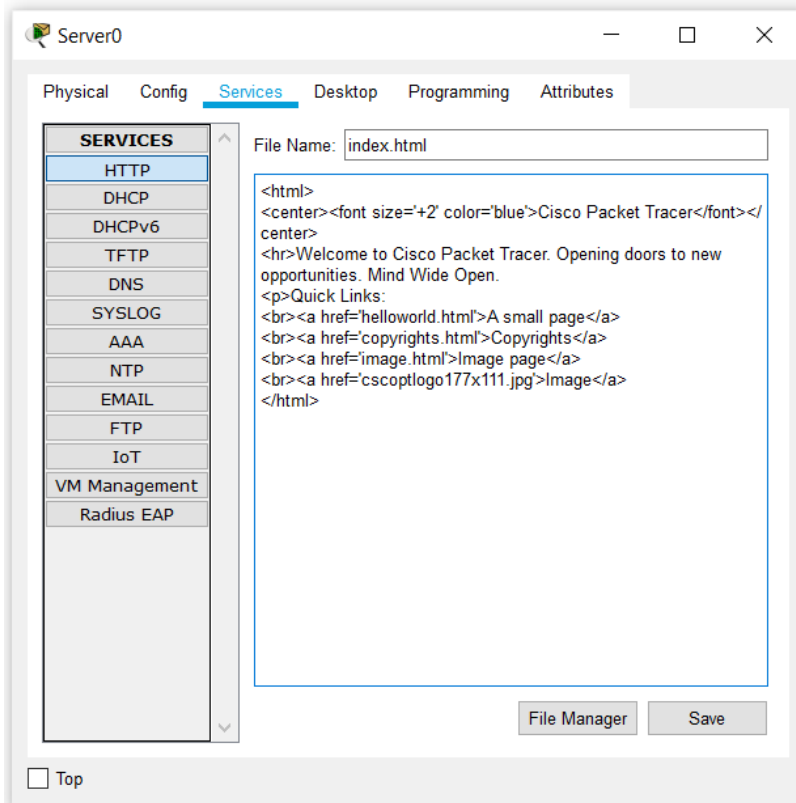
HTTPS ☒ On ☐ Off

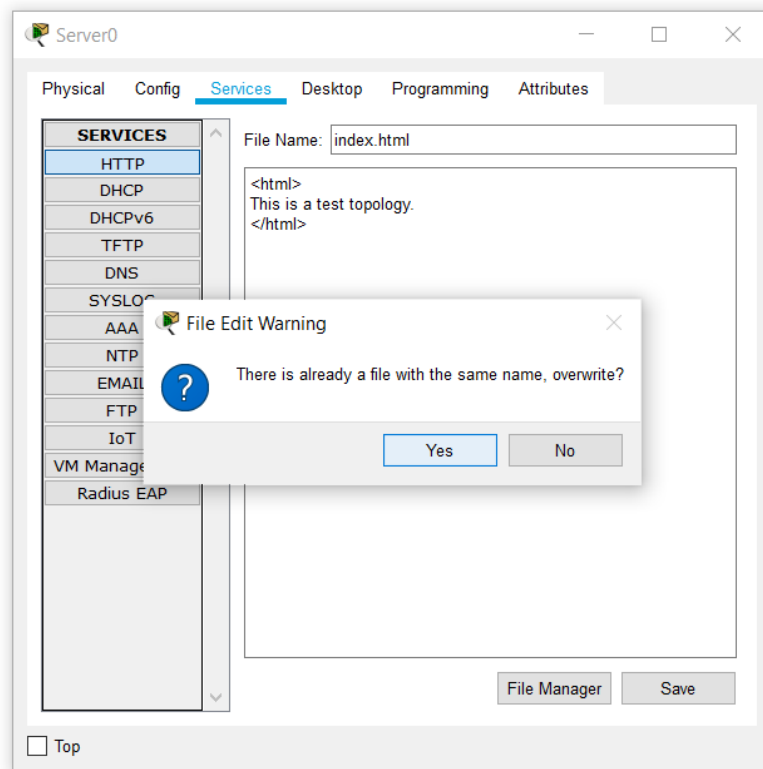
File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscptlogo177x11...		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

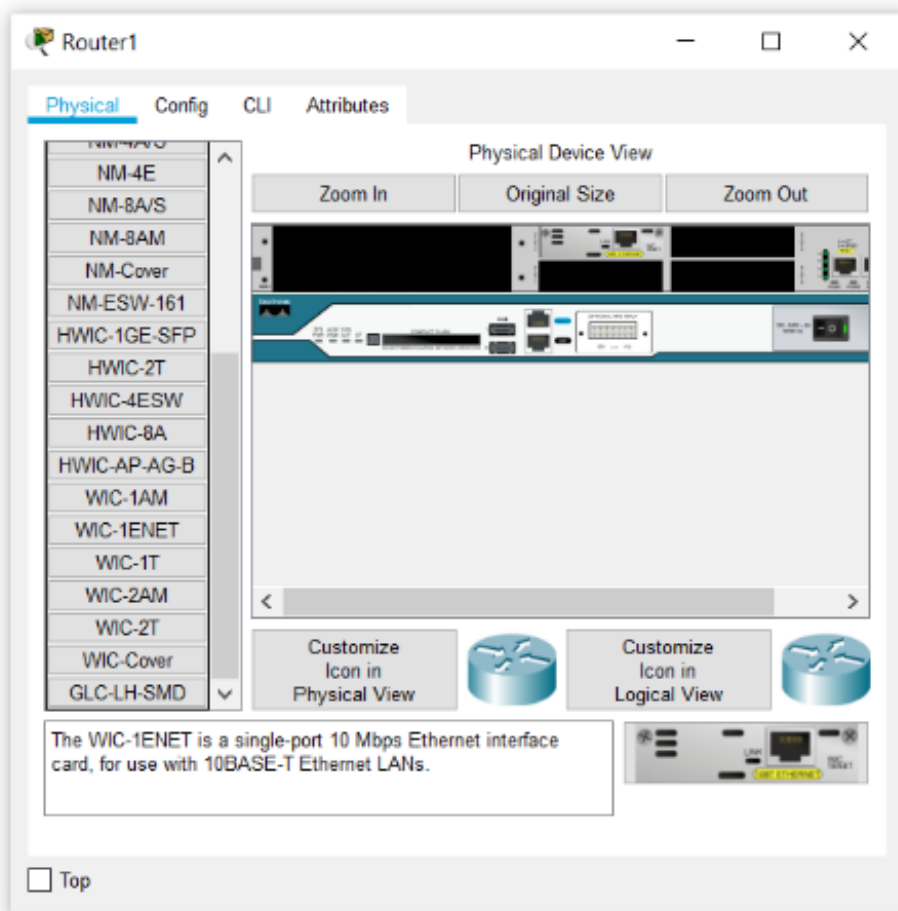
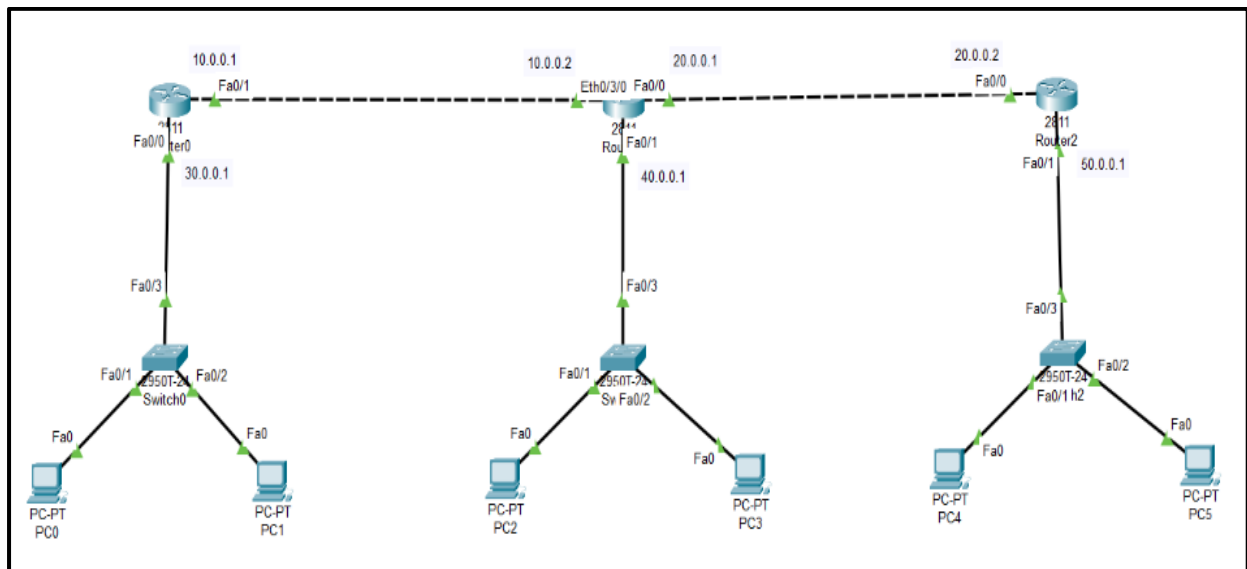
New FileImport

☐ Top





EIGRP



Router-0

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 30.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#interface fastethernet 0/1

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#exit

Router(config)#router eigrp ?

<1-65535> Autonomous system number

```
Router(config)#router eigrp 100
Router(config-router)#network 0.0.0.0
Router(config-router)#exit
Router(config)#
```

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
Router(config-if)#exit
Router(config)#interface fastethernet 0/1
Router(config-if)#ip address 40.0.0.1 255.0.0.0
Router(config-if)#no shutdown
```

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
```

Router(config-if)#**exit**

Router(config)#**interface ethernet 0/3/0**

Router(config-if)#**ip address 10.0.0.2 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface Ethernet0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/3/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**router eigrp ?**

<1-65535> Autonomous system number

Router(config)#**router eigrp 100**

Router(config-router)#**network 0.0.0.0**

Router(config-router)#**exit**

Router(config)#

Router-2

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 20.0.0.2 255.0.0.0**

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#interface fastethernet 0/1

Router(config-if)#ip address 50.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#exit

Router(config)#router eigrp ?

<1-65535> Autonomous system number

Router(config)#router eigrp 100

Router(config-router)#network 0.0.0.0

Router(config-router)#exit

Router(config)#

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:5CFF:FEE3:7772

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:58FF:FE2A:B7C8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::205:2AFF:FE86:C9E1

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FE19:C410

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC4

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE5A:8358

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC5

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:43FF:FE1E:BDBE

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Physical Config Desktop Programming Attributes

Command Prompt

Packet Tracer PC Command Line 1.0

C:\>ping 50.0.0.3

Pinging 50.0.0.3 with 32 bytes of data:

Request timed out.

Reply from 50.0.0.3: bytes=32 time=1ms TTL=125

Reply from 50.0.0.3: bytes=32 time=1ms TTL=125

Reply from 50.0.0.3: bytes=32 time=1ms TTL=125

Ping statistics for 50.0.0.3:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 50.0.0.3

Pinging 50.0.0.3 with 32 bytes of data:

Reply from 50.0.0.3: bytes=32 time<1ms TTL=125

Reply from 50.0.0.3: bytes=32 time<1ms TTL=125

Reply from 50.0.0.3: bytes=32 time<1ms TTL=125

Reply from 50.0.0.3: bytes=32 time=1ms TTL=125

Ping statistics for 50.0.0.3:

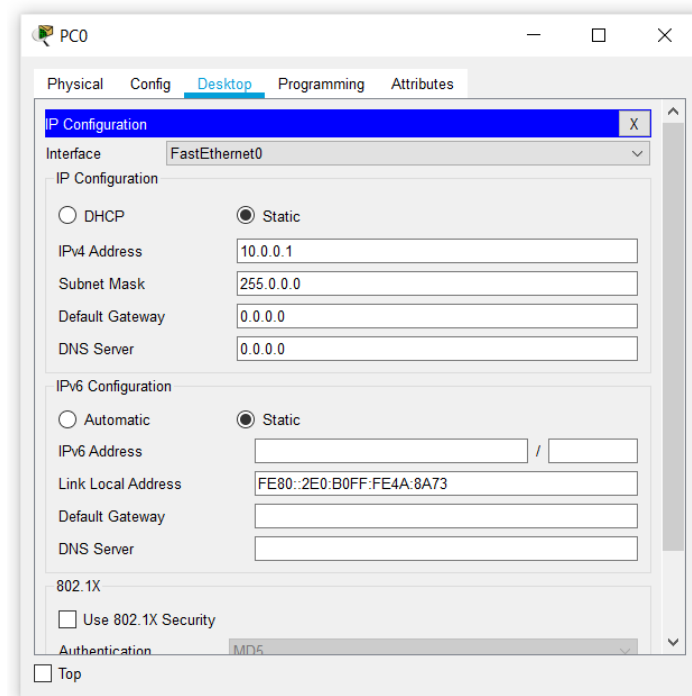
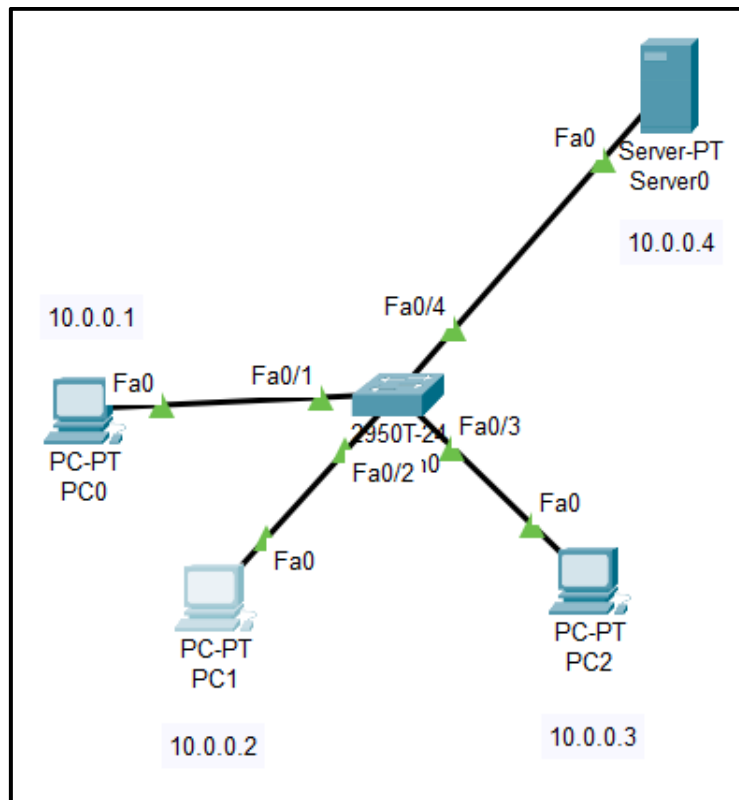
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|

Port Security



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::204:9AFF:FEC0:97CA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FE4C:1970

Default Gateway

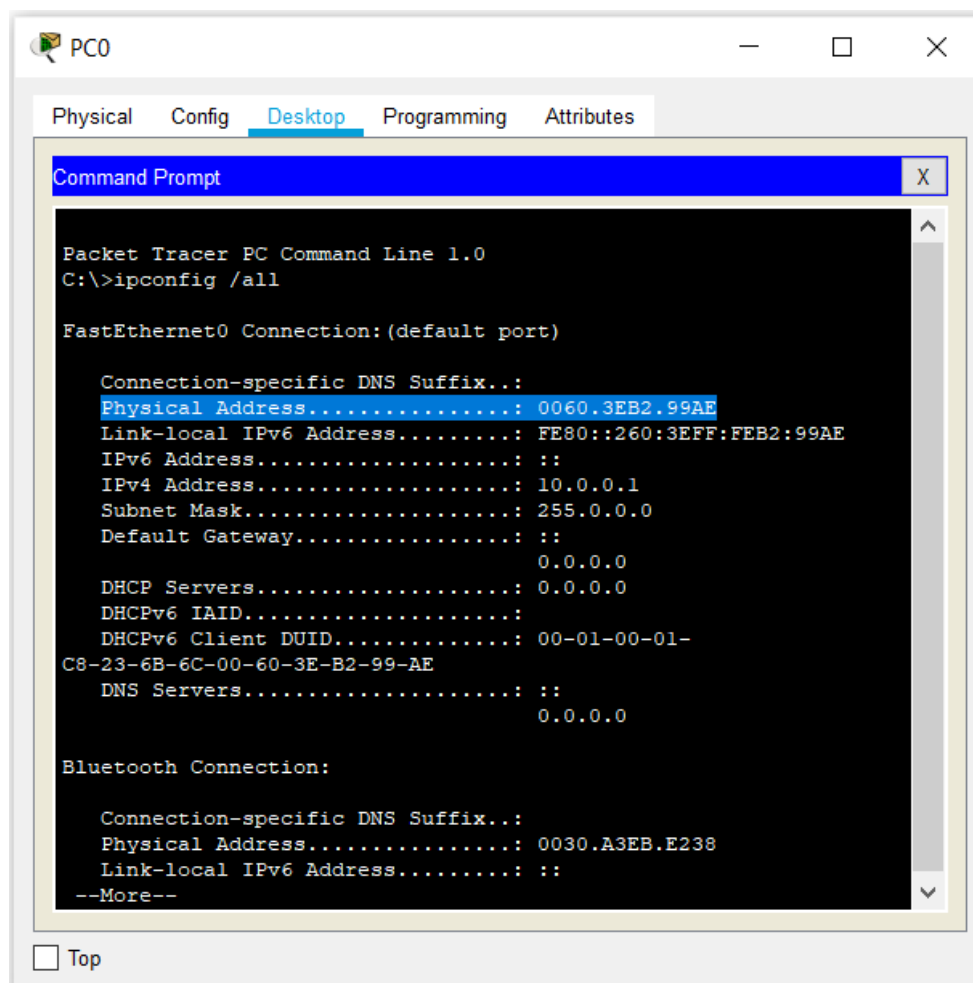
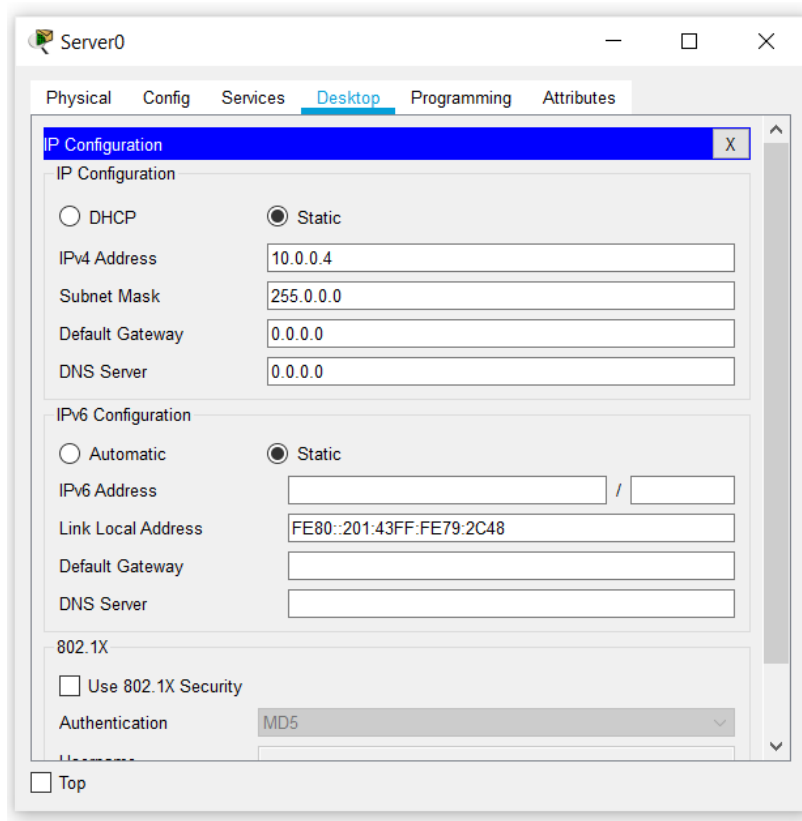
DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top



Switch>enable

Switch#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#interface fastethernet 0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport port-security

Switch(config-if)#switchport port-security ?

aging Port-security aging commands

mac-address Secure mac address

maximum Max secure addresses

violation Security violation mode

<cr>

Switch(config-if)#switchport port-security mac-address 0060.3EB2.99AE

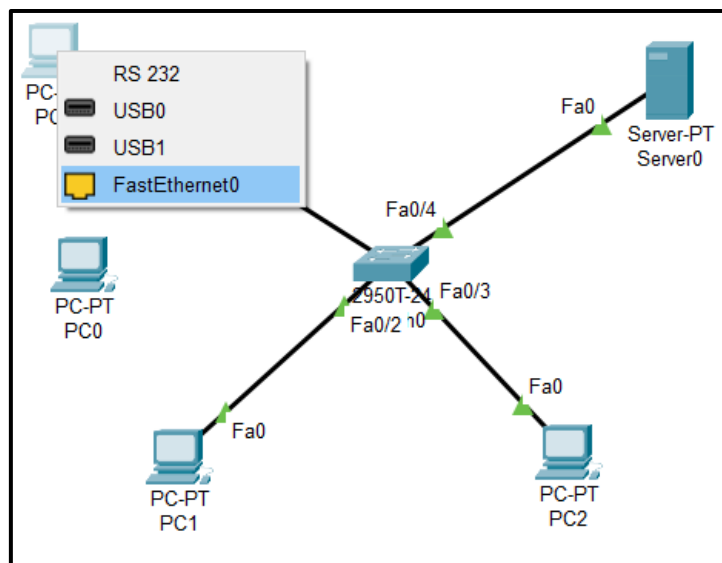
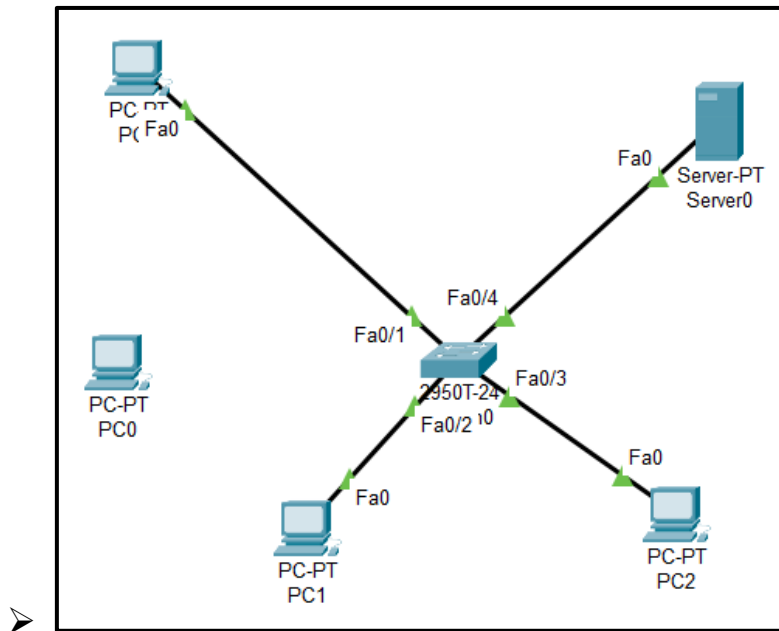
Switch(config-if)#switchport port-security maximum 1

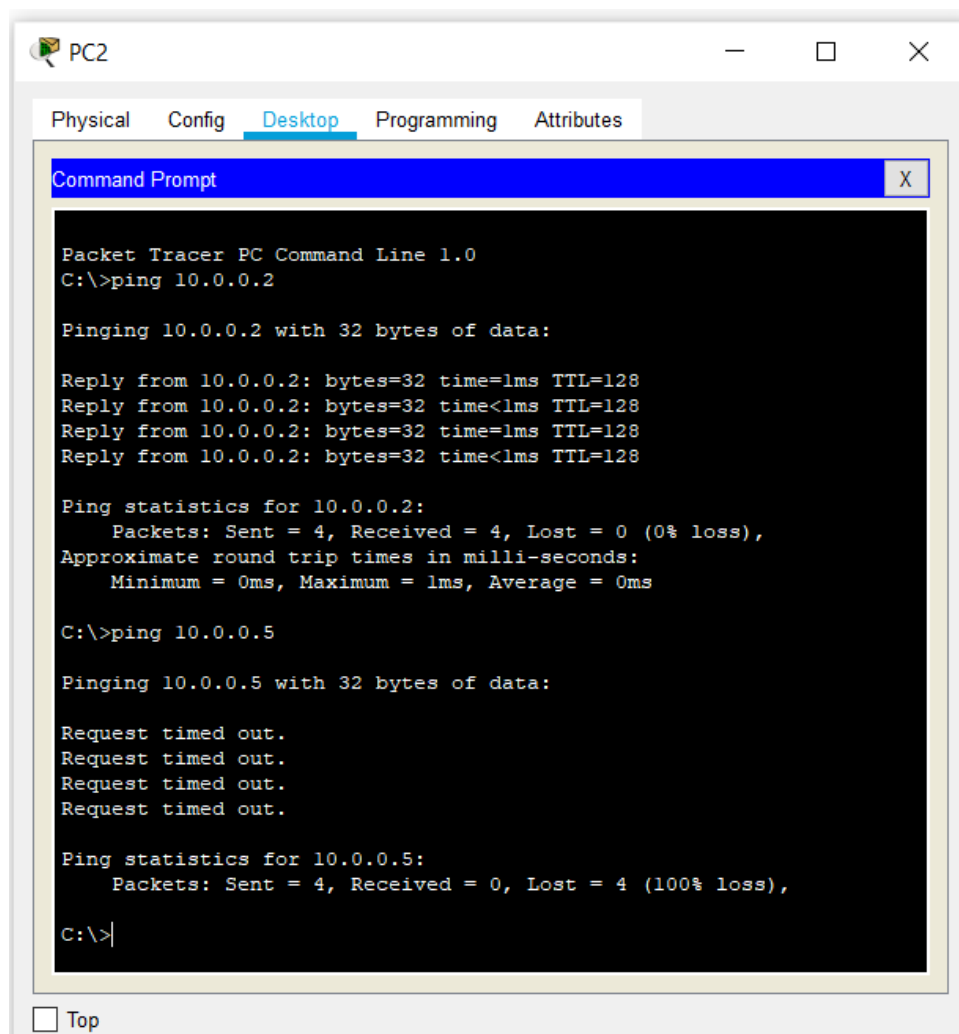
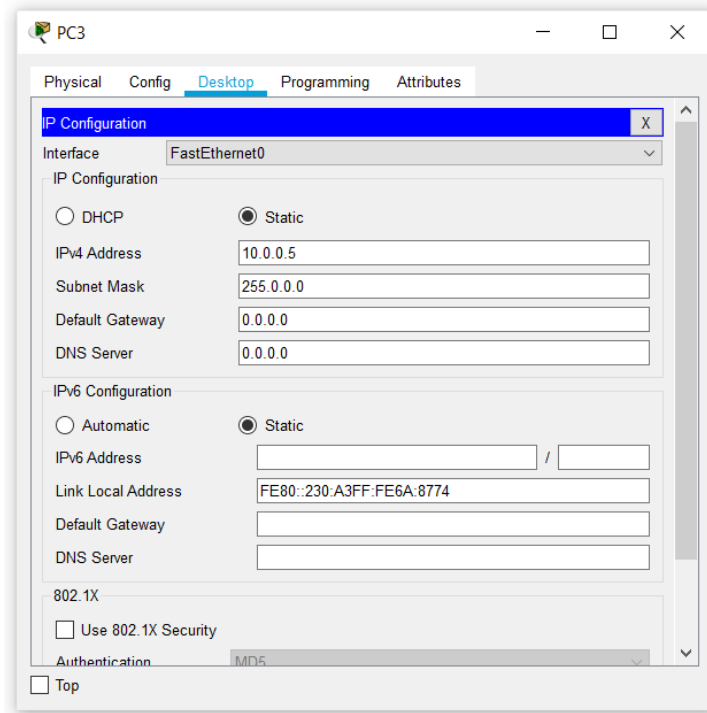
Switch(config-if)#switchport port-security violation shutdown

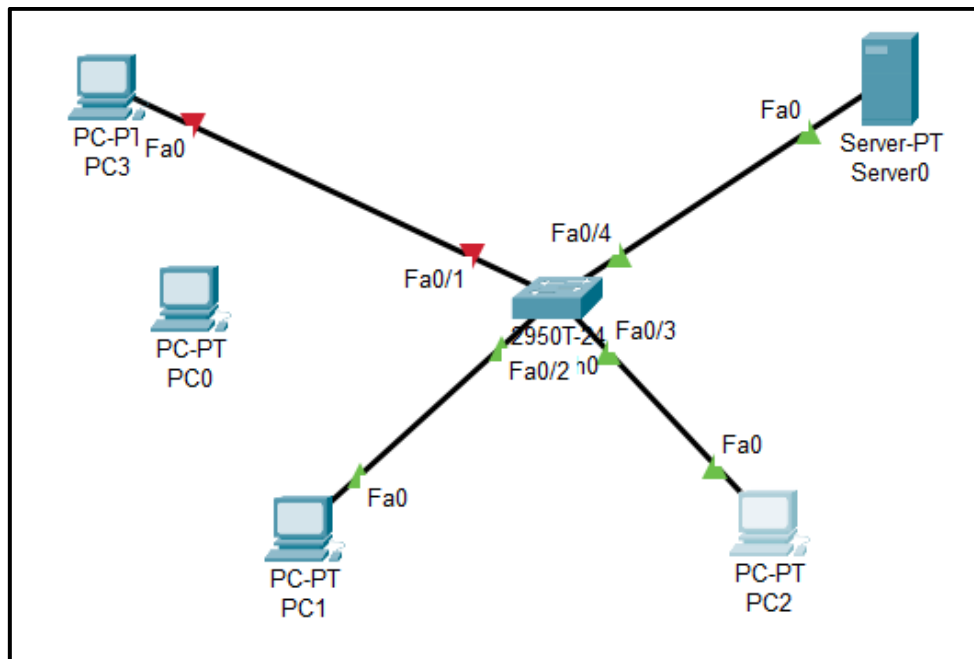
Switch(config-if)#exit

Switch(config)#

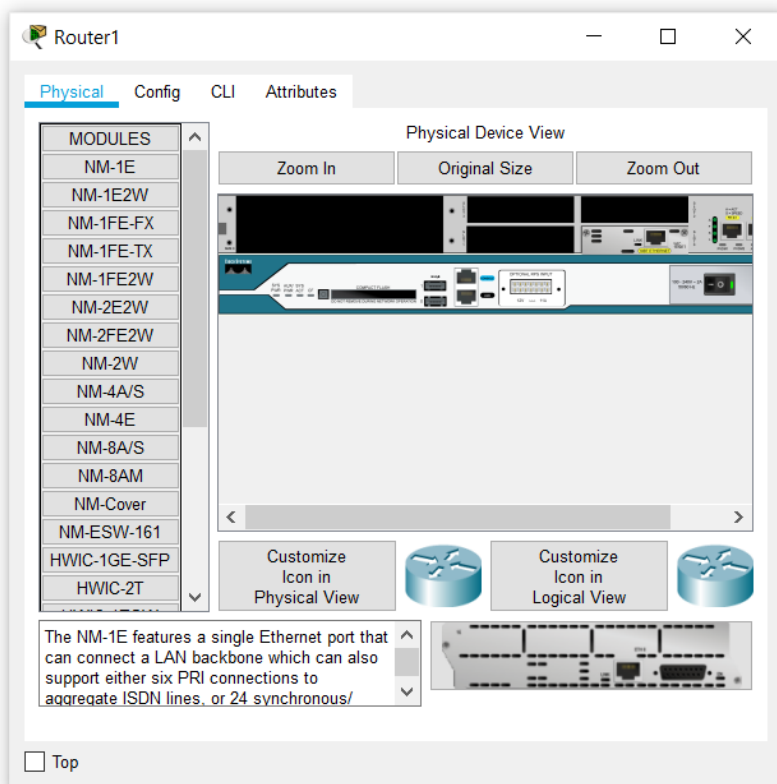
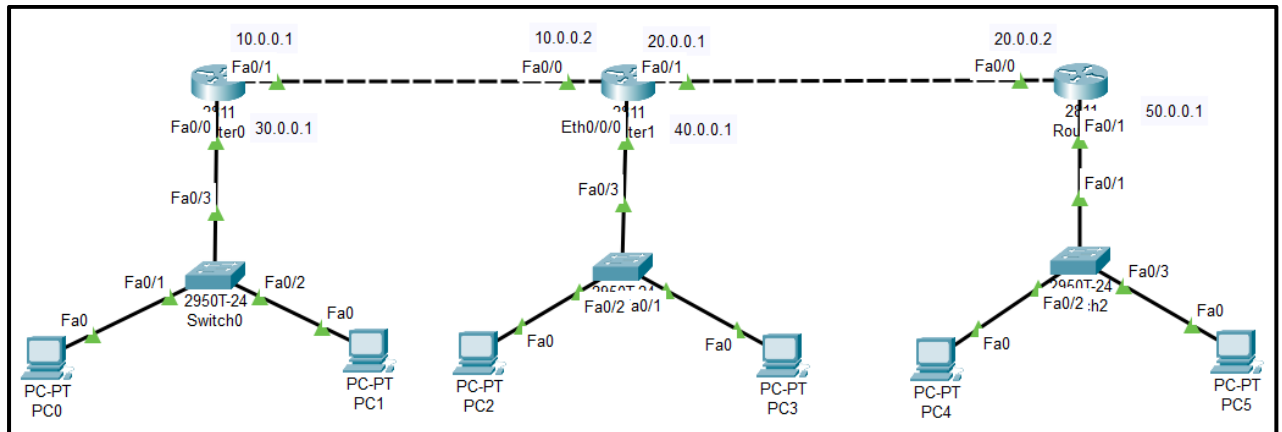
- If the cable is disconnected and plugged in an unauthorized device, after pinging the unauthorized device, the connection gets lost.







ACL



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:70FF:FEC8:26B9

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:D3FF:FEAB:59D3

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE9E:42D8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FEE8:197A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:43FF:FEC5:9EB8

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:A3FF:FEB2:11BD

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Router-0

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 30.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#**exit**

Router(config)#**router eigrp ?**

<1-65535> Autonomous system number

Router(config)#**router eigrp 100**


```
Router(config-router)#network 0.0.0.0
```

```
Router(config-router)#exit
```

```
Router(config)#
```

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
```

```
Router#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#interface fastethernet 0/0
```

```
Router(config-if)#ip address 10.0.0.2 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

```
Router(config-if)#exit
```

```
Router(config)#interface fastethernet 0/1
```

```
Router(config-if)#ip address 20.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

```
Router(config-if)#exit
```

Router(config)#**interface ethernet 0/0/0**

Router(config-if)#**ip address 40.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface Ethernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**router eigrp ?**

<1-65535> Autonomous system number

Router(config)#**router eigrp 100**

Router(config-router)#**network 0.0.0.0**

Router(config-router)#

%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 10.0.0.1

(FastEthernet0/0) is up: new adjacency

Router(config-router)#**exit**

Router(config)#

Router-2

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

```
Router(config-if)#ip address 20.0.0.2 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,  
changed state to up
```

```
Router(config-if)#exit
```

```
Router(config)#interface fastethernet 0/1
```

```
Router(config-if)#ip address 50.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,  
changed state to up
```

```
Router(config-if)#exit
```

```
Router(config)#router eigrp 100
```

```
Router(config-router)#network 0.0.0.0
```

```
Router(config-router)#
```

```
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 20.0.0.1  
(FastEthernet0/0) is up: new adjacency
```

```
Router(config-router)#exit
```

```
Router(config)#access-list ?
```

<1-99> IP standard access list

<100-199> IP extended access list

Router(config)#**access-list 10 ?**

deny Specify packets to reject

permit Specify packets to forward

remark Access list entry comment

Router(config)#**access-list 10 deny ?**

A.B.C.D Address to match

any Any source host

host A single host address

Router(config)#**access-list 10 deny host ?**

A.B.C.D Host address

Router(config)#**access-list 10 deny host 50.0.0.1 ?**

<cr>

Router(config)#**access-list 10 permit any**

Router(config)#**exit**

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip access-group ?**

<1-199> IP access list (standard or extended)

WORD Access-list name

Router(config-if)#**ip access-group 10 ?**

in inbound packets

out outbound packets

Router(config-if)#**ip access-group 10 in**

Router(config-if)#exit

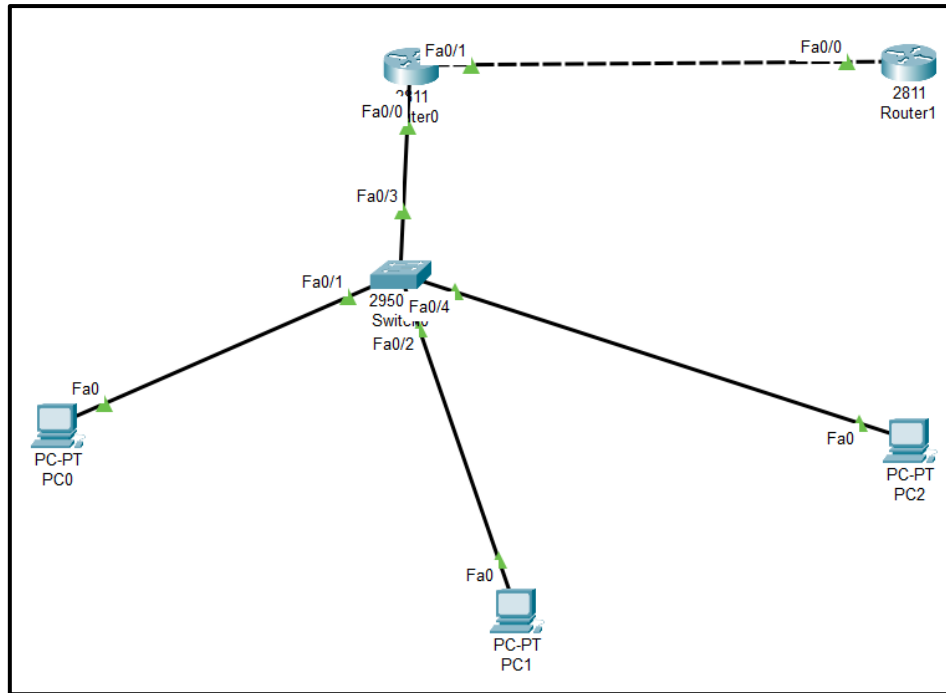
Router(config)#interface fastethernet 0/1

Router(config-if)#ip access-group 10 out

Router(config-if)#exit

Router(config)#

NAT



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FEBE:6613

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:A3FF:FE02:72C2

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.4

Subnet Mask 255.0.0.0

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:70FF:FECD:3B20

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Router-0

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 20.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#exit

Router(config)#router eigrp ?

<1-65535> Autonomous system number

Router(config)#router eigrp 100

Router(config-router)#network 0.0.0.0

Router(config-router)#exit

Router(config)#ip nat inside source static 10.0.0.2 20.0.0.10

Router(config)#interface fastethernet 0/1

Router(config-if)#ip nat inside

Router(config-if)#exit

Router(config)#interface fastethernet 0/0

Router(config-if)#ip nat outside

Router(config-if)#exit

Router(config)#

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 20.0.0.2 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#**exit**

Router(config)#

Router(config)#**router eigrp ?**

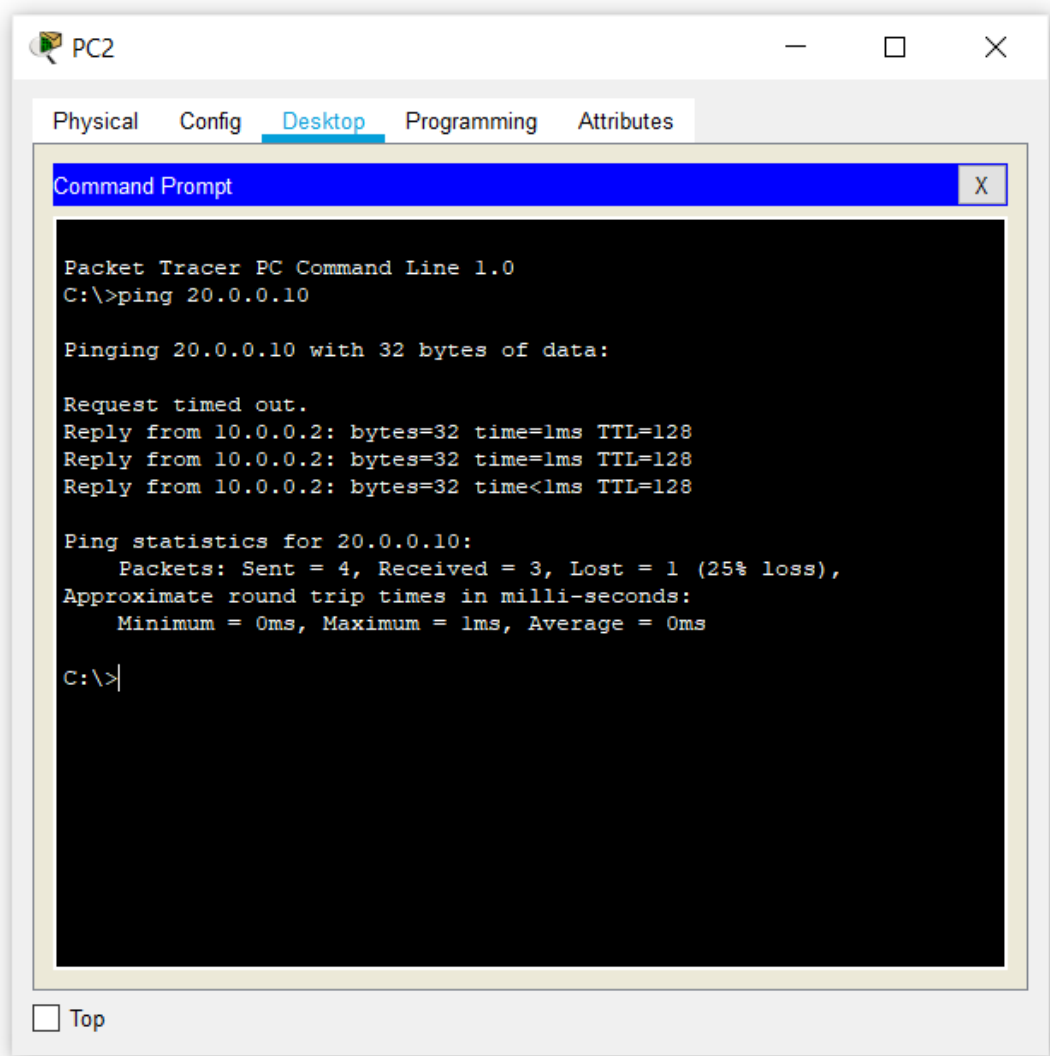
<1-65535> Autonomous system number

Router(config)#**router eigrp 100**

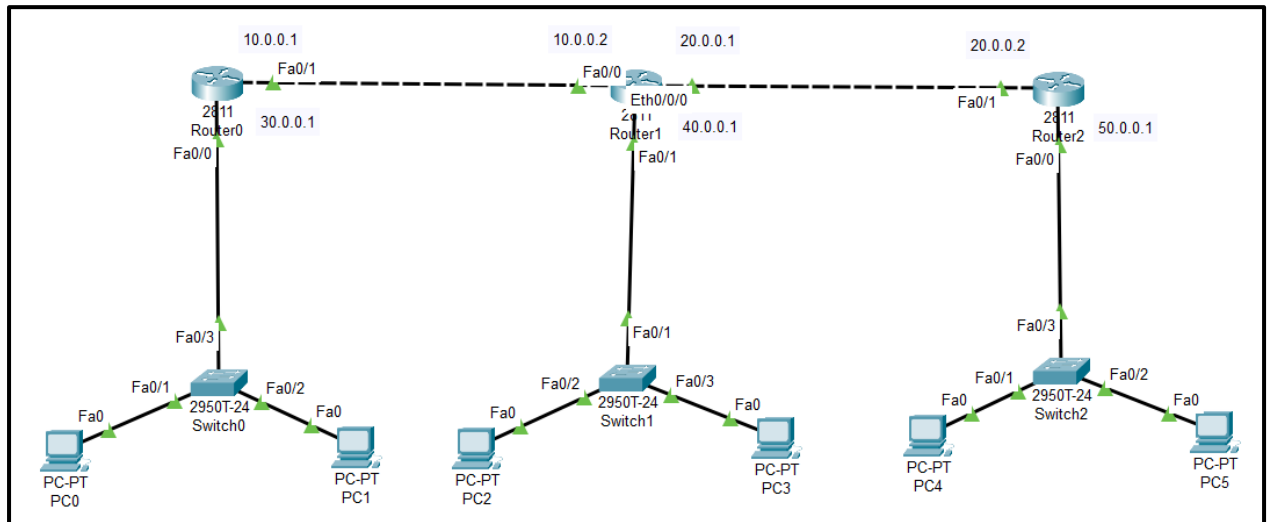
Router(config-router)#**network 0.0.0.0**

Router(config-router)#**exit**

Router(config)#



OSPF



Router1

Physical

Config

CLI

Attributes

NM-4V-S

NM-4E

NM-8A/S

NM-8AM

NM-Cover

NM-ESW-161

HWIC-1GE-SFP

HWIC-2T

HWIC-4ESW

HWIC-8A

HWIC-AP-AG-B

WIC-1AM

WIC-1ENET

WIC-1T

WIC-2AM

WIC-2T

WIC-Cover

GLC-LH-SMD

Physical Device View

Zoom In

Original Size

Zoom Out



Customize Icon in Physical View

Customize Icon in Logical View

The WIC-1ENET is a single-port 10 Mbps Ethernet interface card, for use with 10BASE-T Ethernet LANs.



☐ Top

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F9FF:FE23:DC7C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FEAC:BD43

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE8A:1220

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::230:F2FF:FEAC:D00D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:58FF:FEC7:2593

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FE64:C611

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Router-0

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 30.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#**exit**

Router(config)#**router ospf ?**

<1-65535> Process ID

Router(config)#**router ospf 100**


```
Router(config-router)#network 10.0.0.0 255.0.0.0 area 1
```

```
Router(config-router)#network 30.0.0.0 255.0.0.0 area 1
```

```
Router(config-router)#exit
```

```
Router(config)#
```

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
```

```
Router#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#interface fastethernet 0/0
```

```
Router(config-if)#ip address 10.0.0.2 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

```
Router(config-if)#exit
```

```
Router(config)#interface fastethernet 0/1
```

```
Router(config-if)#ip address 40.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface ethernet 0/0/0**

Router(config-if)#**ip address 20.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface Ethernet0/0/0, changed state to up

Router(config-if)#**exit**

Router(config)#**router ospf 100**

Router(config-router)#**network 10.0.0.0 255.0.0.0 area 1**

Router(config-router)#**network 20.0.0.0 255.0.0.0 area 1**

00:14:51: %OSPF-5-ADJCHG: Process 100, Nbr 30.0.0.1 on FastEthernet0/0
from LOADING to FULL, Loading Done

Router(config-router)#**network 40.0.0.0 255.0.0.0 area 1**

Router(config-router)#**exit**

Router(config)#

Router-2

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

```
Router(config-if)#ip address 50.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,  
changed state to up
```

```
Router(config-if)#exit
```

```
Router(config)#interface fastethernet 0/1
```

```
Router(config-if)#ip address 20.0.0.2 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,  
changed state to up
```

```
Router(config-if)#exit
```

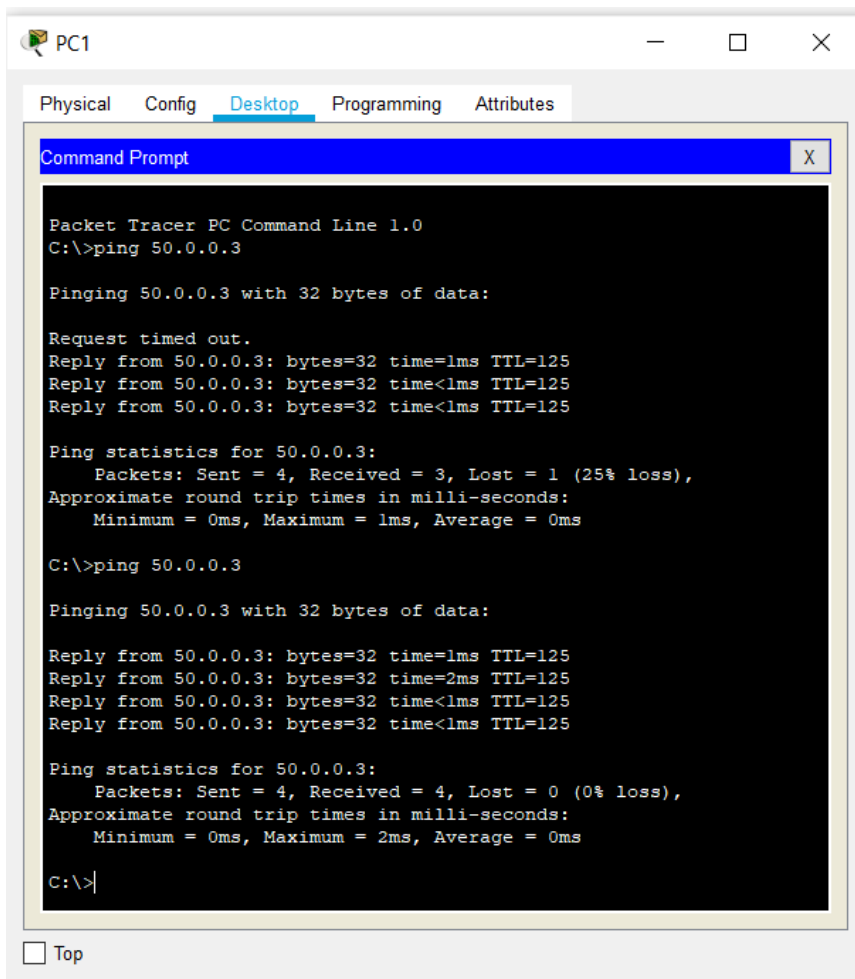
```
Router(config)#router ospf 100
```

```
Router(config-router)#network 20.0.0.0 255.0.0.0 area 1
```

```
Router(config-router)#network 50.0.0.0 255.0.0.0 area 1
```

```
Router(config-router)#exit
```

```
Router(config)#
```



BGP

Anonymous number:

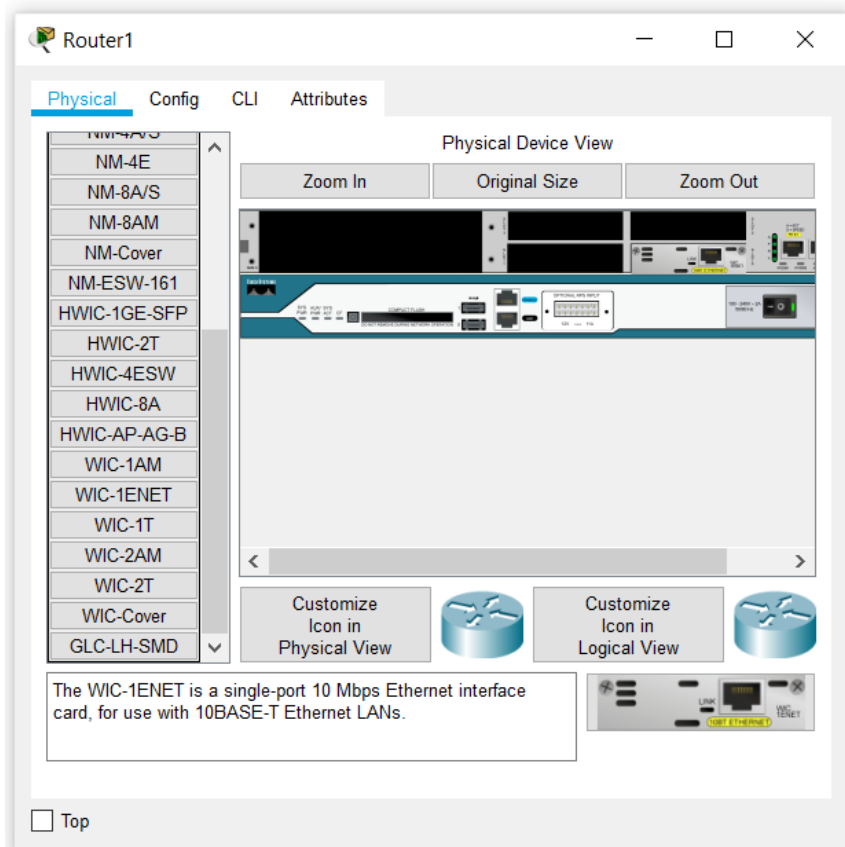
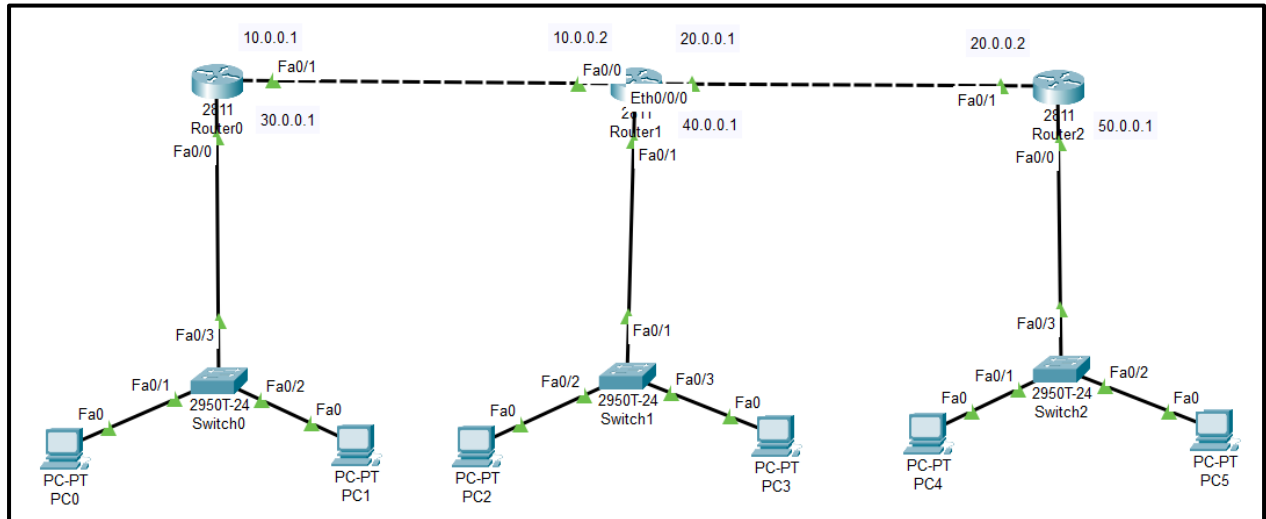
100

Anonymous number:

200

Anonymous number:

300



PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F9FF:FE23:DC7C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 30.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 30.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FEAC:BD43

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE8A:1220

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC3

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 40.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 40.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::230:F2FF:FEAC:D00D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC4

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.2

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:58FF:FEC7:2593

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC5

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 50.0.0.3

Subnet Mask 255.0.0.0

Default Gateway 50.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FE64:C611

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

Router-0

Would you like to enter the initial configuration dialog? [yes/no]: **no**

Router>**enable**

Router#**configure terminal**

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#**interface fastethernet 0/0**

Router(config-if)#**ip address 30.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#**exit**

Router(config)#**interface fastethernet 0/1**

Router(config-if)#**ip address 10.0.0.1 255.0.0.0**

Router(config-if)#**no shutdown**

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

Router(config-if)#**exit**

Router(config)#**router bgp 100**

Router(config-router)#**network 10.0.0.0 mask 255.0.0.0**

Router(config-router)#**network 30.0.0.0 mask 255.0.0.0**

```
Router(config-router)#neighbor 10.0.0.2 remote-as 200
```

```
Router(config-router)#exit
```

```
Router(config)#
```

Router-1

Would you like to enter the initial configuration dialog? [yes/no]: no

```
Router>enable
```

```
Router#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#interface fastethernet 0/0
```

```
Router(config-if)#ip address 20.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

```
Router(config-if)#exit
```

```
Router(config)#interface fastethernet 0/1
```

```
Router(config-if)#ip address 40.0.0.1 255.0.0.0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

```
Router(config-if)#exit
```

Router(config)#interface ethernet 0/0/0

Router(config-if)#ip address 10.0.0.2 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface Ethernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0/0,
changed state to up

Router(config-if)#exit

Router(config)#router bgp 200

Router(config-router)#network 10.0.0.0 mask 255.0.0.0

Router(config-router)#network 20.0.0.0 mask 255.0.0.0

Router(config-router)#network 40.0.0.0 mask 255.0.0.0

Router(config-router)#neighbor 10.0.0.1 remote-as 100

Router(config-router)#%BGP-5-ADJCHANGE: neighbor 10.0.0.1 Up

Router(config-router)#neighbor 20.0.0.2 remote-as 300

Router(config-router)#exit

Router(config)#

Router-2

Would you like to enter the initial configuration dialog? [yes/no]: no

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastethernet 0/0

Router(config-if)#ip address 50.0.0.1 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

Router(config-if)#exit

Router(config)#interface fastethernet 0/1

Router(config-if)#ip address 20.0.0.2 255.0.0.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

Router(config-if)#exit

Router(config)#router bgp 300

Router(config-router)#network 20.0.0.0 mask 255.0.0.0

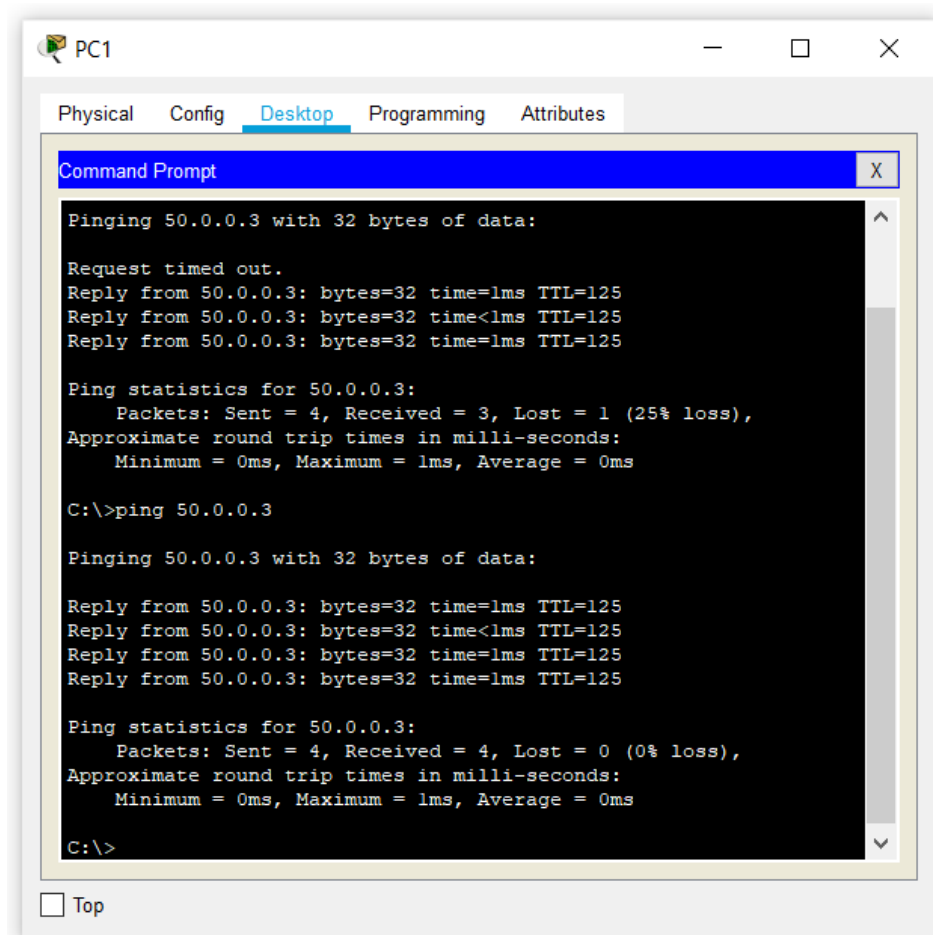
Router(config-router)#network 50.0.0.0 mask 255.0.0.0

Router(config-router)#neighbor 20.0.0.1 remote-as 200

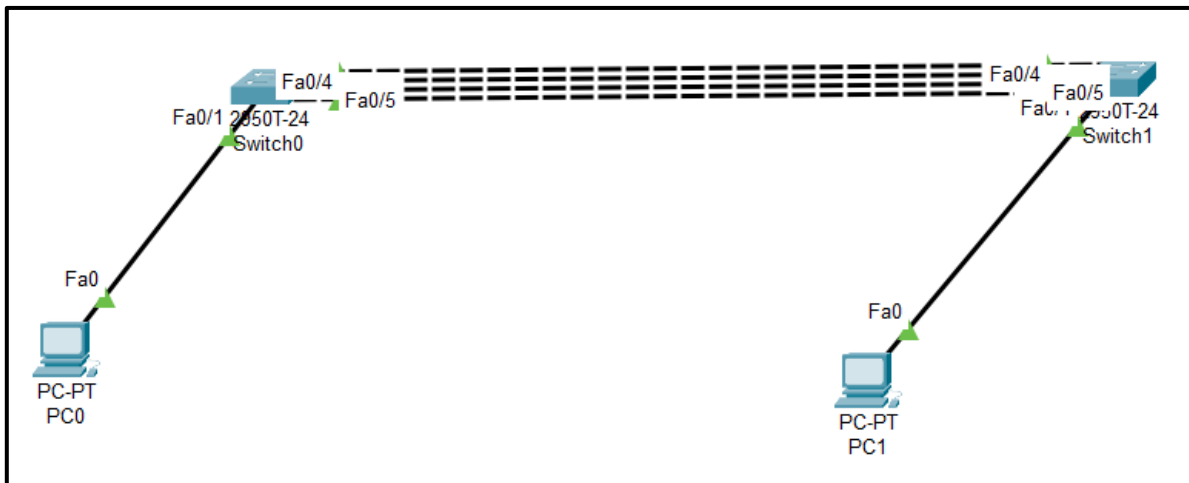
Router(config-router)#%BGP-5-ADJCHANGE: neighbor 20.0.0.1 Up

Router(config-router)#exit

Router(config)#



Etherchannel



Switch-0

Switch>enable

Switch#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0001.6412.2DCB

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0001.6412.2DCB

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/4 Desg FWD 19 128.4 P2p

Fa0/5 Desg FWD 19 128.5 P2p

Fa0/1 Desg FWD 19 128.1 P2p

Fa0/2 Desg FWD 19 128.2 P2p

Fa0/3 Desg FWD 19 128.3 P2p

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#**interface range fa 0/2, fa 0/3, fa0/4, fa 0/5**

Switch(config-if-range)#**channel-group ?**

<1-6> Channel group number

Switch(config-if-range)#**channel-group 1 mode on**

Switch(config-if-range)#

Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1,
changed state to up

Switch(config-if-range)#**exit**

Switch(config)#**exit**

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#**show spanning-tree**

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0001.6412.2DCB

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0001.6412.2DCB

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Po1 Desg FWD 7 128.27 Shr

Fa0/1 Desg FWD 19 128.1 P2p

Switch#

Switch-1

Switch>enable

Switch#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0001.6412.2DCB

Cost 19

Port 2(FastEthernet0/2)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0090.2BD5.292D

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/2 Root FWD 19 128.2 P2p

Fa0/1 Desg FWD 19 128.1 P2p

Fa0/4 Desg FWD 19 128.4 P2p

Fa0/3 Desg FWD 19 128.3 P2p

Fa0/5 Desg FWD 19 128.5 P2p

Switch#[configure terminal](#)

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#[interface range fa 0/2, fa 0/3, fa 0/4, fa 0/5](#)

Switch(config-if-range)#[channel-group 1 mode on](#)

Switch(config-if-range)#

Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1,
changed state to up

Switch(config-if-range)#[exit](#)

Switch(config)#[exit](#)

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#[show spanning-tree](#)

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0001.6412.2DCB

Cost 7

Port 27(Port-channel1)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0090.2BD5.292D

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface Role Sts Cost Prio.Nbr Type

Fa0/1 Desg FWD 19 128.1 P2p

Po1 Root LSN 7 128.27 Shr

Switch#